



Penalty System Documentation

Scope: Complete documentation of the reachability-constrained penalty system, including Mach trajectory guidance, lever penalty guidance, and their integration with 3D Dynamic Programming for optimal climb trajectory generation.

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1) System Overview and Objectives

1.1 Purpose and Scope

The penalty system implements a sophisticated guidance framework that uses soft constraints to guide trajectory optimization toward realistic, operationally viable solutions. The system employs reachability-constrained Mach guidance and engine-friendly lever penalties to create balanced, cooperative optimization that produces smooth, realistic climb trajectories.

1.2 System Objectives

Primary Objectives:

- **Trajectory Guidance:** Guide optimization toward realistic Mach trajectories
- **Engine Protection:** Prevent excessive lever usage and engine wear
- **Operational Viability:** Ensure generated trajectories are operationally feasible
- **Balanced Optimization:** Create cooperative penalty interaction for smooth guidance

Key Components:

- Reachability-constrained Mach guidance system
- Engine-friendly lever penalty system
- Adaptive penalty weighting with altitude progress
- Integration with 3D Dynamic Programming
- Cooperative penalty interaction and balancing

1.3 System Flow Overview

The penalty system follows a logical progression:

1. **Mach Trajectory Analysis:** Calculate reachability constraints based on achievable Mach rates
 2. **Penalty Calculation:** Compute Mach and lever penalties for current state
 3. **Adaptive Weighting:** Adjust penalty weights based on altitude progress and urgency
 4. **Cost Integration:** Combine penalties with base fuel cost for total optimization cost
 5. **Guidance Application:** Apply penalties during 3D Dynamic Programming optimization
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2) Mathematical Foundation

2.1 Penalty System Philosophy

Theory: Instead of hard constraints that can create infeasible solutions, the penalty system uses graduated costs that increase with deviation from desired behavior. This approach provides guidance while maintaining optimization flexibility.

Mathematical Formulation:

$$J_{total} = J_{fuel} + J_{mach_penalty} + J_{lever_penalty}$$

Where:

- $J_{\text{fuel}} = \dot{m}/P_s$ (original fuel cost density)
- $J_{\text{mach_penalty}}$ (Mach guidance penalty)
- $J_{\text{lever_penalty}}$ (lever penalty)

Physical Interpretation:

- **Base fuel cost:** Primary optimization objective
- **Mach penalty:** Guidance toward realistic trajectories
- **Lever penalty:** Engine protection and operational limits
- **Cooperative interaction:** Penalties work together for balanced guidance

2.2 Reachability-Constrained Approach

Theory: The system creates dynamic "safety corridors" that ensure the target remains achievable throughout the climb using realistic Mach change rates, rather than forcing predetermined trajectories.

Key Innovation: Dynamic corridor bounds based on achievable Mach rates rather than artificial trajectory assumptions.

Mathematical Foundation:

$$\text{Remaining_Steps} = (1.0 - \text{Altitude_Fraction}) \times \text{Total_Steps}$$

$$\text{Max_Mach_Change} = \text{Remaining_Steps} \times \text{Max_Mach_Rate}$$

$$\text{Safety_Corridor} = [\text{Target_Mach} - \text{Max_Mach_Change}, \text{Target_Mach} + \text{Max_Mach_Change}]$$

2.3 Adaptive Penalty Weighting

Theory: Penalty weights increase with altitude progress to create urgency and ensure convergence to target conditions.

Mathematical Formulation:

$$w_{\text{penalty}} = w_{\text{base}} \times (1.0 + \alpha \times \text{Altitude_Fraction})$$

Where:

- `w_{base}` = Base penalty weight
- `\alpha` = Urgency multiplier
- `\text{Altitude}\_Fraction` = Current progress (0.0 to 1.0)

3) System Architecture and Data Structures

3.1 System Parameters

Mach Trajectory Guidance Parameters:

```
MACH_TRAJECTORY_GUIDANCE = True           # Enable Mach guidance
MACH_PENALTY_BASE_WEIGHT = 0.3             # Base penalty weight
URGENCY_MULTIPLIER = 2.0                  # Urgency scaling factor
GUIDANCE_PENALTY_WEIGHT = 0.5             # Guidance penalty inside corridor
MAX_REASONABLE_MACH_RATE = 0.02           # Maximum realistic Mach change per step
TOTAL_CLIMB_STEPS_ESTIMATE = 50            # Estimated total climb steps
TARGET_MACH_TOLERANCE = 0.015             # Target Mach tolerance
```

Lever Penalty Parameters:

```
LEVER_PENALTY_GUIDANCE = True             # Enable lever penalties
LEVER_PENALTY_WEIGHT = 3.0                 # Base lever penalty weight
LEVER_PENALTY_THRESHOLD = 0.85             # Threshold above which penalties apply
LEVER_PENALTY_EXPONENT = 3.0               # Exponent for penalty curve
LEVER_PENALTY_CRITICAL_THRESHOLD = 0.90    # Critical threshold
LEVER_PENALTY_CRITICAL_MULTIPLIER = 5.0    # Critical range multiplier
LEVER_PENALTY_ULTRA_CRITICAL_THRESHOLD = 0.95 # Ultra-critical threshold
LEVER_PENALTY_ULTRA_CRITICAL_MULTIPLIER = 20.0 # Ultra-critical multiplier
```

3.2 Data Structures

Penalty Calculation State:

```
@dataclass
class PenaltyState:
    current_mach: float
    target_mach: float
    altitude_fraction: float
    current_lever: float
    total_steps_estimate: int
    max_mach_rate: float
```

Penalty Results:

```
@dataclass
class PenaltyResults:
    mach_penalty: float
    lever_penalty: float
    total_penalty: float
    penalty_weight: float
    guidance_active: bool
    corridor_bounds: Tuple[float, float]
```

4) Mach Trajectory Guidance System

4.1 Reachability Constraint Calculation

Function: `compute_mach_penalty()`

Purpose: Calculate Mach trajectory guidance penalty based on reachability constraints.

Algorithm:

```

def compute_mach_penalty(current_mach: float, target_mach: float,
                        prev_mach: float = None,
                        altitude_fraction: float = None) -> float:
    """
    Calculate Mach trajectory guidance penalty using reachability constraints.

    Args:
        current_mach: Current Mach number
        target_mach: Target Mach number at cruise altitude
        prev_mach: Previous Mach number (for rate calculation)
        altitude_fraction: Current altitude progress (0.0 to 1.0)

    Returns:
        Mach penalty value
    """

    if not MACH_TRAJECTORY_GUIDANCE:
        return 0.0

    # Calculate remaining steps
    remaining_steps = (1.0 - altitude_fraction) * TOTAL_CLIMB_STEPS_ESTIMATE

    # Calculate maximum achievable Mach change
    max_mach_change = remaining_steps * MAX_REASONABLE_MACH_RATE

    # Create safety corridor
    corridor_low = target_mach - max_mach_change
    corridor_high = target_mach + max_mach_change

    # Check if current Mach is within safety corridor
    if corridor_low <= current_mach <= corridor_high:
        # Inside corridor - apply guidance penalty
        mach_deviation = abs(current_mach - target_mach)
        penalty_weight = GUIDANCE_PENALTY_WEIGHT

        # Scale penalty with altitude progress (urgency)
        urgency_multiplier = 1.0 + URGENCY_MULTIPLIER * altitude_fraction
        penalty_weight *= urgency_multiplier

```

```

        return penalty_weight * (mach_deviation ** 2)
    else:
        # Outside corridor - apply strong penalty
        mach_deviation = min(abs(current_mach - corridor_low),
                             abs(current_mach - corridor_high))

        penalty_weight = MACH_PENALTY_BASE_WEIGHT

        # Scale penalty with altitude progress (urgency)
        urgency_multiplier = 1.0 + URGENCY_MULTIPLIER * altitude_fraction
        penalty_weight *= urgency_multiplier

    return penalty_weight * (mach_deviation ** 2)

```

4.2 Dynamic Corridor Management

Corridor Calculation:

```

def calculate_safety_corridor(target_mach: float, altitude_fraction: float,
                             total_steps_estimate: int, max_mach_rate: float) -> Tuple[float, float]:
    """Calculate dynamic safety corridor bounds."""

    # Calculate remaining steps
    remaining_steps = (1.0 - altitude_fraction) * total_steps_estimate

    # Calculate maximum achievable Mach change
    max_mach_change = remaining_steps * max_mach_rate

    # Create corridor bounds
    corridor_low = target_mach - max_mach_change
    corridor_high = target_mach + max_mach_change

    return corridor_low, corridor_high

```

4.3 Adaptive Urgency Scaling

Urgency Calculation:

```
def calculate_urgency_weight(base_weight: float, altitude_fraction: float,
                             urgency_multiplier: float = 2.0) -> float:
    """Calculate adaptive penalty weight based on altitude progress."""

    return base_weight * (1.0 + urgency_multiplier * altitude_fraction)
```

5) Lever Penalty Guidance System

5.1 Lever Penalty Calculation

Function: `compute_lever_penalty()`

Purpose: Calculate engine-friendly lever penalties to prevent excessive usage.

Algorithm:

```

def compute_lever_penalty(current_lever: float, altitude_fraction: float = None) -> float:
    """
    Calculate lever position penalty for excessive usage.

    Args:
        current_lever: Current lever position (0.0 to 1.0)
        altitude_fraction: Current altitude progress (0.0 to 1.0)

    Returns:
        Lever penalty value
    """

    if not LEVER_PENALTY_GUIDANCE:
        return 0.0

    # No penalty below threshold
    if current_lever <= LEVER_PENALTY_THRESHOLD:
        return 0.0

    # Calculate excess above threshold
    excess = current_lever - LEVER_PENALTY_THRESHOLD

    # Base penalty with cubic growth
    penalty = LEVER_PENALTY_WEIGHT * (excess ** LEVER_PENALTY_EXPONENT)

    # Apply critical range multipliers
    if current_lever >= LEVER_PENALTY_ULTRA_CRITICAL_THRESHOLD:
        penalty *= LEVER_PENALTY_ULTRA_CRITICAL_MULTIPLIER
    elif current_lever >= LEVER_PENALTY_CRITICAL_THRESHOLD:
        penalty *= LEVER_PENALTY_CRITICAL_MULTIPLIER

    # Scale with altitude progress (urgency)
    if altitude_fraction is not None:
        urgency_multiplier = 1.0 + URGENCY_MULTIPLIER * altitude_fraction
        penalty *= urgency_multiplier

    return penalty

```

5.2 Multi-Tier Penalty Structure

Penalty Tiers:

1. **Safe Range** (0.0 - 0.85): No penalty
2. **Warning Range** (0.85 - 0.90): Base penalty with cubic growth
3. **Critical Range** (0.90 - 0.95): 5x multiplier
4. **Ultra-Critical Range** (0.95 - 1.0): 20x multiplier

Mathematical Formulation:

$$\text{Penalty} = \begin{cases} 0 & \text{if } \ell \leq 0.85 \\ w \times (\ell - 0.85)^3 & \text{if } 0.85 < \ell \leq 0.90 \\ 5w \times (\ell - 0.85)^3 & \text{if } 0.90 < \ell \leq 0.95 \\ 20w \times (\ell - 0.85)^3 & \text{if } \ell > 0.95 \end{cases}$$

Where w is the base penalty weight.

6) Penalty Integration and Balancing

6.1 Total Cost Calculation

Function: `calculate_total_penalty_cost()`

Purpose: Integrate all penalties with base fuel cost for total optimization cost.

Implementation:

```
def calculate_total_penalty_cost(base_fuel_cost: float, current_mach: float,
                                target_mach: float, current_lever: float,
                                altitude_fraction: float, prev_mach: float = None) -> float:
    """Calculate total cost including all penalties."""

    # Calculate individual penalties
    mach_penalty = compute_mach_penalty(current_mach, target_mach, prev_mach, altitude_fraction)
    lever_penalty = compute_lever_penalty(current_lever, altitude_fraction)

    # Calculate total penalty
    total_penalty = mach_penalty + lever_penalty

    # Calculate total cost
    total_cost = base_fuel_cost + total_penalty

    return total_cost
```

6.2 Cooperative Penalty Interaction

Theory: The system ensures penalties work together rather than against each other, creating balanced guidance.

Interaction Principles:

1. **Mach guidance** encourages realistic trajectories
2. **Lever penalties** protect engine operation
3. **Adaptive weighting** ensures convergence
4. **Cooperative balancing** prevents conflicting guidance

6.3 Penalty Weight Balancing

Weight Management:

```
def balance_penalty_weights(mach_penalty: float, lever_penalty: float,
                           altitude_fraction: float) -> Tuple[float, float]:
    """Balance penalty weights to ensure cooperative interaction."""

    # Calculate relative magnitudes
    total_penalty = mach_penalty + lever_penalty

    if total_penalty > 0:
        mach_weight = mach_penalty / total_penalty
        lever_weight = lever_penalty / total_penalty

    # Ensure balanced influence
    if mach_weight > 0.8: # Mach penalty too dominant
        mach_penalty *= 0.8
        lever_penalty *= 1.2
    elif lever_weight > 0.8: # Lever penalty too dominant
        lever_penalty *= 0.8
        mach_penalty *= 1.2

    return mach_penalty, lever_penalty
```

7) Code Execution Flow and Logic

7.1 System Entry Point and Initialization

Main Entry Function: Integration with 3D Dynamic Programming

Execution Sequence:

```
# 1. Penalty System Initialization
def initialize_penalty_system():
    """Initialize penalty system parameters and state."""

    # Set up penalty parameters
    penalty_params = {
        'mach_guidance': MACH_TRAJECTORY_GUIDANCE,
        'lever_guidance': LEVER_PENALTY_GUIDANCE,
        'base_weights': {
            'mach': MACH_PENALTY_BASE_WEIGHT,
            'lever': LEVER_PENALTY_WEIGHT
        },
        'thresholds': {
            'lever': LEVER_PENALTY_THRESHOLD,
            'critical': LEVER_PENALTY_CRITICAL_THRESHOLD,
            'ultra_critical': LEVER_PENALTY_ULTRA_CRITICAL_THRESHOLD
        }
    }

    return penalty_params
```

7.2 Penalty Calculation Flow

Function: `calculate_penalties_for_state()`

Step-by-Step Execution Flow:

```
# PHASE 1: STATE ANALYSIS
```

```
def calculate_penalties_for_state(current_mach: float, current_lever: float,
                                  altitude_fraction: float, target_mach: float,
                                  prev_mach: float = None):

    # Step 1: Calculate altitude progress
    altitude_progress = altitude_fraction if altitude_fraction is not None else 0.0

    # Step 2: Determine remaining climb steps
    remaining_steps = (1.0 - altitude_progress) * TOTAL_CLIMB_STEPS_ESTIMATE

    # Step 3: Calculate maximum achievable Mach change
    max_mach_change = remaining_steps * MAX_REASONABLE_MACH_RATE
```

```
# PHASE 2: MACH PENALTY CALCULATION
```

```
    # Step 4: Calculate safety corridor bounds
    corridor_low = target_mach - max_mach_change
    corridor_high = target_mach + max_mach_change

    # Step 5: Determine penalty zone
    if corridor_low <= current_mach <= corridor_high:
        # Inside corridor - guidance penalty
        mach_deviation = abs(current_mach - target_mach)
        penalty_weight = GUIDANCE_PENALTY_WEIGHT
        guidance_active = True
    else:
        # Outside corridor - strong penalty
        mach_deviation = min(abs(current_mach - corridor_low),
                             abs(current_mach - corridor_high))
        penalty_weight = MACH_PENALTY_BASE_WEIGHT
        guidance_active = False

    # Step 6: Apply urgency scaling
    urgency_multiplier = 1.0 + URGENCY_MULTIPLIER * altitude_progress
    penalty_weight *= urgency_multiplier

    # Step 7: Calculate Mach penalty
    mach_penalty = penalty_weight * (mach_deviation ** 2)
```

```

# PHASE 3: LEVER PENALTY CALCULATION
# Step 8: Check lever threshold
if current_lever <= LEVER_PENALTY_THRESHOLD:
    lever_penalty = 0.0
else:
    # Step 9: Calculate excess above threshold
    excess = current_lever - LEVER_PENALTY_THRESHOLD

    # Step 10: Base penalty with cubic growth
    lever_penalty = LEVER_PENALTY_WEIGHT * (excess ** LEVER_PENALTY_EXPONENT)

    # Step 11: Apply critical range multipliers
    if current_lever >= LEVER_PENALTY_ULTRA_CRITICAL_THRESHOLD:
        lever_penalty *= LEVER_PENALTY_ULTRA_CRITICAL_MULTIPLIER
    elif current_lever >= LEVER_PENALTY_CRITICAL_THRESHOLD:
        lever_penalty *= LEVER_PENALTY_CRITICAL_MULTIPLIER

    # Step 12: Apply urgency scaling
    lever_penalty *= urgency_multiplier

```

```

# PHASE 4: PENALTY INTEGRATION
# Step 13: Calculate total penalty
total_penalty = mach_penalty + lever_penalty

# Step 14: Create penalty results
penalty_results = PenaltyResults(
    mach_penalty=mach_penalty,
    lever_penalty=lever_penalty,
    total_penalty=total_penalty,
    penalty_weight=penalty_weight,
    guidance_active=guidance_active,
    corridor_bounds=(corridor_low, corridor_high)
)

return penalty_results

```

7.3 Integration with 3D Dynamic Programming

Function: `integrate_penalties_with_dp()`

Integration Flow:

```
def integrate_penalties_with_dp(base_cost: float, mach: float, lever: float,
                                altitude_fraction: float, target_mach: float) -> float:
    """Integrate penalties with 3D DP cost calculation."""

    # Calculate penalties for current state
    penalty_results = calculate_penalties_for_state(
        current_mach=mach,
        current_lever=lever,
        altitude_fraction=altitude_fraction,
        target_mach=target_mach
    )

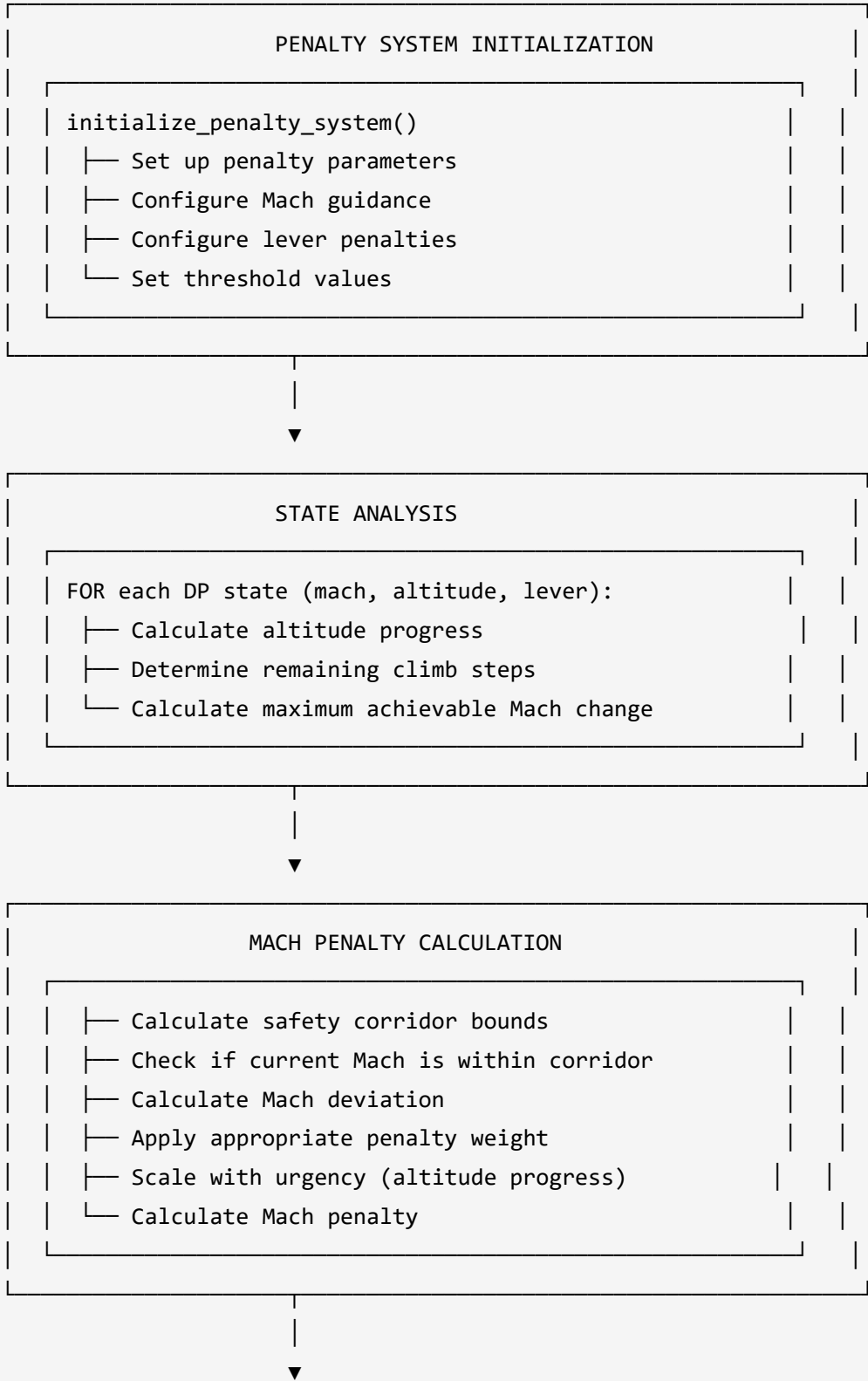
    # Add penalties to base fuel cost
    total_cost = base_cost + penalty_results.total_penalty

    return total_cost
```

7.4 Visual Code Flow Diagram

PENALTY SYSTEM EXECUTION FLOW

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LEVER PENALTY CALCULATION

- └─ Check lever threshold
- └─ Calculate excess above threshold
- └─ Apply cubic penalty growth
- └─ Apply critical range multipliers
- └─ Scale with urgency (altitude progress)
- └─ Calculate lever penalty



PENALTY INTEGRATION

- └─ Sum Mach and lever penalties
- └─ Add to base fuel cost
- └─ Balance penalty weights
- └─ Return total optimization cost



RETURN TOTAL COST

Cost = Base Fuel Cost + Total Penalties

7.5 Function Call Hierarchy

```
3D Dynamic Programming Optimization
├─ compute_3d_cost()
│   ├─ calculate_base_fuel_cost()
│   └─ calculate_total_penalty_cost()
│       ├─ compute_mach_penalty()
│       │   ├─ calculate_safety_corridor()
│       │   └─ calculate_urgency_weight()
│       │       └─ determine_penalty_zone()
│       └─ compute_lever_penalty()
│           ├─ check_lever_threshold()
│           └─ calculate_excess()
│               └─ apply_critical_multipliers()
│                   └─ apply_urgency_scaling()
│                       └─ balance_penalty_weights()
└─ integrate_penalties_with_dp()
    ├─ calculate_penalties_for_state()
    ├─ create_penalty_results()
    └─ return_total_cost()
```

8) Integration and Interface

8.1 3D Dynamic Programming Integration

Function: `integrate_with_3d_dp()`

Integration Process:

```

def integrate_with_3d_dp():
    """Integrate penalty system with 3D Dynamic Programming optimization."""

    # Modify the cost calculation in DP
    def enhanced_cost_calculation(mach: float, lever: float, altitude_fraction: float,
                                   base_cost: float, target_mach: float = 0.7) -> float:
        """Enhanced cost calculation with penalty integration."""

        # Calculate penalties
        penalty_results = calculate_penalties_for_state(
            current_mach=mach,
            current_lever=lever,
            altitude_fraction=altitude_fraction,
            target_mach=target_mach
        )

        # Return total cost
        return base_cost + penalty_results.total_penalty

    return enhanced_cost_calculation

```

8.2 Main Interface Functions

Penalty System Interface:

```

def run_penalty_guided_optimization(aero: AeroTables, eng: EngineWrapper,
                                    M_grid: np.ndarray, H_sched: np.ndarray,
                                    target_mach: float = 0.7) -> MinFuelSchedule:
    """Run 3D DP optimization with penalty guidance."""

    # Initialize penalty system
    penalty_params = initialize_penalty_system()

    # Enhanced cost function with penalties
    def penalty_enhanced_cost(mach: float, altitude_m: float, lever: float,
                              mass_kg: float) -> float:
        # Calculate base fuel cost
        base_cost = compute_3d_cost(aero, eng, mach, altitude_m, lever, mass_kg)

        # Calculate altitude fraction
        altitude_fraction = (altitude_m - H_sched[0]) / (H_sched[-1] - H_sched[0])

        # Add penalties
        penalty_cost = calculate_total_penalty_cost(
            base_cost, mach, target_mach, lever, altitude_fraction
        )

        return penalty_cost

    # Run 3D DP with penalty-enhanced cost
    dp_result = ClimbingCore.DynamicProgrammingOptimizer.solve_3d_fixed_mass(
        aero=aero, eng=eng, M_grid=M_grid, H_sched=H_sched,
        lever_samples=50, target_mach=target_mach
    )

    return dp_result

```

8.3 Configuration Interface

Penalty Configuration:

```
def configure_penalty_system(mach_guidance: bool = True, lever_guidance: bool = True,
                             target_mach: float = 0.7, urgency_multiplier: float = 2.0) -> Dict[str, float]:
    """Configure penalty system parameters."""

    config = {
        'mach_guidance_enabled': mach_guidance,
        'lever_guidance_enabled': lever_guidance,
        'target_mach': target_mach,
        'urgency_multiplier': urgency_multiplier,
        'parameters': {
            'mach_base_weight': MACH_PENALTY_BASE_WEIGHT,
            'lever_base_weight': LEVER_PENALTY_WEIGHT,
            'lever_threshold': LEVER_PENALTY_THRESHOLD,
            'max_mach_rate': MAX_REASONABLE_MACH_RATE
        }
    }

    return config
```

9) Validation and Quality Assurance

9.1 Penalty System Validation

Parameter Validation:

```

def validate_penalty_parameters() -> None:
    """Validate penalty system parameters."""

    # Validate Mach guidance parameters
    if MACH_PENALTY_BASE_WEIGHT < 0:
        raise ValueError("Mach penalty base weight must be positive")

    if URGENCY_MULTIPLIER < 0:
        raise ValueError("Urgency multiplier must be positive")

    if MAX_REASONABLE_MACH_RATE <= 0:
        raise ValueError("Maximum Mach rate must be positive")

    # Validate lever penalty parameters
    if LEVER_PENALTY_WEIGHT < 0:
        raise ValueError("Lever penalty weight must be positive")

    if not (0.0 <= LEVER_PENALTY_THRESHOLD <= 1.0):
        raise ValueError("Lever penalty threshold must be between 0.0 and 1.0")

    if LEVER_PENALTY_EXPONENT <= 0:
        raise ValueError("Lever penalty exponent must be positive")

```

9.2 Performance Monitoring

Penalty Effectiveness Monitoring:

```

def monitor_penalty_effectiveness(dp_result: MinFuelSchedule, target_mach: float) -> Dict[str, float]:
    """Monitor effectiveness of penalty guidance."""

    # Analyze Mach trajectory
    mach_trajectory = dp_result.mach
    mach_deviations = np.abs(mach_trajectory - target_mach)
    max_mach_deviation = np.max(mach_deviations)
    avg_mach_deviation = np.mean(mach_deviations)

    # Analyze lever usage
    lever_trajectory = dp_result.lever
    max_lever = np.max(lever_trajectory)
    avg_lever = np.mean(lever_trajectory)
    high_lever_fraction = np.sum(lever_trajectory > LEVER_PENALTY_THRESHOLD) / len(lever_trajectory)

    # Calculate effectiveness metrics
    effectiveness_metrics = {
        'max_mach_deviation': max_mach_deviation,
        'avg_mach_deviation': avg_mach_deviation,
        'mach_convergence': 1.0 - (max_mach_deviation / target_mach),
        'max_lever_usage': max_lever,
        'avg_lever_usage': avg_lever,
        'high_lever_fraction': high_lever_fraction,
        'lever_protection': 1.0 - high_lever_fraction
    }

    return effectiveness_metrics

```

9.3 Error Handling

Penalty Calculation Errors:

```

def safe_penalty_calculation(current_mach: float, current_lever: float,
                             altitude_fraction: float, target_mach: float) -> PenaltyResults:
    """Safe penalty calculation with error handling."""

    try:
        # Validate inputs
        if not (0.0 <= current_lever <= 1.0):
            raise ValueError(f"Invalid lever position: {current_lever}")

        if not (0.0 <= altitude_fraction <= 1.0):
            raise ValueError(f"Invalid altitude fraction: {altitude_fraction}")

        # Calculate penalties
        penalty_results = calculate_penalties_for_state(
            current_mach, current_lever, altitude_fraction, target_mach
        )

        return penalty_results

    except Exception as e:
        print(f"Penalty calculation error: {e}")
        # Return zero penalties as fallback
        return PenaltyResults(
            mach_penalty=0.0,
            lever_penalty=0.0,
            total_penalty=0.0,
            penalty_weight=0.0,
            guidance_active=False,
            corridor_bounds=(0.0, 1.0)
        )

```

End of Penalty System Documentation